MODUL 5
ALJABAR BOOLEAN

1. TUJUAN PRAKTIKUM

Mahasiswa mampu mengimplementasikan aljabar *boolean* dalam proyek simulasi.

2. TEORI
2.1. ALJABAR BOOLEAN

Aljabar boolean atau *Boolean algebra* telah menjadi dasar dari perkembangan ilmu komputer dan logika digital. Dalam ilmu matematika dan logika matematika, aljabar boolean merupakan sub area dari aljabar yang mana variable-variablenya bernilai benar (*true*) atau salah (*false*), atau umumnya dinotasikan dengan 1 dan 0. Hubungan logika antar variable biner dalam aljabar boolean menggunakan tiga operasi dasar yaitu AND (*conjunction*), OR (*disjunction*), dan NOT (*negation*). Teori dan penjelasan berdasarkan logika AND ditunjukkan pada Tabel 1.

Tabel 1. Teori dan penjelasan logika AND

Theorem		**Explanation**	**Example**
1.	A ● 0 = 0	Any input "AND"ed with 0 = 0, since both inputs in an AND gate must be 1 for the result to be 1	1 ● 0 = 0 0 ● 0 = 0
2.	A ● 1 = A	Any input "AND"ed with 1 = the original input	1 ● 1 = 1 0 ● 1 = 0
3.	A ● A = A	Any input "AND"ed with itself = itself	1 ● 1 = 1 0 ● 0 = 0
4.	A ● $\bar{A}$ = 0	Any input "AND"ed with the opposite of that input will always be zero.	1 ● 0 = 0 0 ● 1 = 0
5.	A ● B = B ● A	Commutative Law - the order of inputs to an AND gate is irrelevant.	
6.	A ● B ● C = (A ● B) ● C = A ● (B ● C)	Associative Law - the order in which successive ANDs are performed is irrelevant.	

Contoh 1

$$Y = (A + B) \bullet (A + B) + C$$
$$= A + B + C \quad \text{(Thm \#3)}$$

Contoh 2

$$
\begin{aligned}
Y &= A \bullet (B + C) \bullet \overline{A} \bullet \overline{B} \bullet D \\
&= A \bullet \overline{A} \bullet (B + C) \bullet \overline{B} \bullet D \quad \text{(Thm \#5)} \\
&= 0 \bullet (B + C) \bullet \overline{B} \bullet D \quad \text{(Thm \#4)} \\
&= 0 \quad \text{(Thm \#1)}
\end{aligned}
$$

Tabel 2. Teori dan penjelasan logika AND

Theorem		Explanation	Example
7.	A + 0 = A	Any input "OR"ed with 0 = the original input.	1 + 0 = 1 0 + 0 = 0
8.	A + 1 = 1	Any input "OR"ed with 1 = 1	1 + 1 = 1 0 + 1 = 1
9.	A + A = A	Any input "OR"ed with itself = itself	1 + 1 = 1 0 + 0 = 0
10.	A + $\bar{A}$ = 1	Any input "OR"ed with the opposite of that input will always be 1.	1 + 0 = 1 0 + 1 = 1
11.	A + B = B + A	Commutative Law - the order of inputs to an OR gate is irrelevant.	
12.	A + B + C = (A + B) + C = A + (B + C)	Associative Law - the order in which successive ORs are performed is irrelevant.	

Contoh 3

$$
\begin{aligned}
Y &= A \bullet \overline{C} + A \bullet \overline{C} \bullet D + 1 \\
&= 1 \quad \text{(Thm \#8)}
\end{aligned}
$$

Contoh 4

$$
\begin{aligned}
Y &= A \bullet B + \overline{A \bullet B} + C \\
&= 1 + C \quad \text{(Thm \#10)} \\
&= 1 \quad \text{(Thm \#8)}
\end{aligned}
$$

Tabel 3. Hukum distribusi

Theorem	Explanation

13.	A ● (B + C) = A ● B + A ● C	Any AND operation can be distributed over an OR operation.
14.	A + (B ● C) = (A + B) ● (A + C)	Any OR operation can be distributed over an AND operation (This does not apply in normal algebra).

Contoh 5

$$
\begin{aligned}
X &= A \bullet B \bullet C + A \bullet C \\
&= A \bullet C \bullet (B+1) && \text{(Thm \#13)} \\
&= A \bullet C && \text{(Thm \#8)}
\end{aligned}
$$

Contoh 6

$$
\begin{aligned}
X &= (A+B) \bullet (A+C) \\
&= A \bullet A + A \bullet C + B \bullet A + B \bullet C && \text{(Thm \#13)} \\
&= A + A \bullet C + B \bullet A + B \bullet C && \text{(Thm \#3)} \\
&= A \bullet (1 + C + B) + B \bullet C && \text{(Thm \#13)} \\
&= A \bullet 1 + B \bullet C && \text{(Thm \#8)} \\
&= A + B \bullet C && \text{(Thm \#2)}
\end{aligned}
$$

Tabel 1. Hukum redudansi

Theorem		**Explanation**
15a.	A + (A ● B) = A	Can be proven using previous theorems: A + (A • B) = (A • 1) + (A • B) (Thm 2) = A • (1 + B) (Thm 13) = A • 1 (Thm 8) = A (Thm 2)
15b.	A ● (A + B) = A	Can be proven using previous theorems: A • (A + B) = (A + 0) • (A + B) (Thm 7) = A + (0 • B) (Thm 14) = A + 0 (Thm 1) = A (Thm 7)
16a.	A + ($\bar{A}$ ● B) = A + B	Can be proven using the previous theorems: A + ($\bar{A}$ • B) = (A + $\bar{A}$) • (A + B) (Thm 14) = 1 • (A + B) (Thm 10) = A + B (Thm 2)

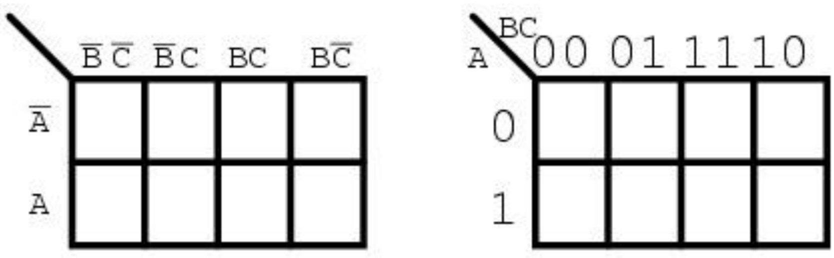
16b.	A ● ($\bar{A}$ + B) = A ● B	Can be proven using the previous theorems: $A \bullet (\bar{A} + B) = (A \bullet \bar{A}) + (A \bullet B)$ (Thm 13) $= 0 + (A \bullet B)$ (Thm 4) $= A \bullet B$ (Thm 7)

Tabel 1. Hukum Demorgan

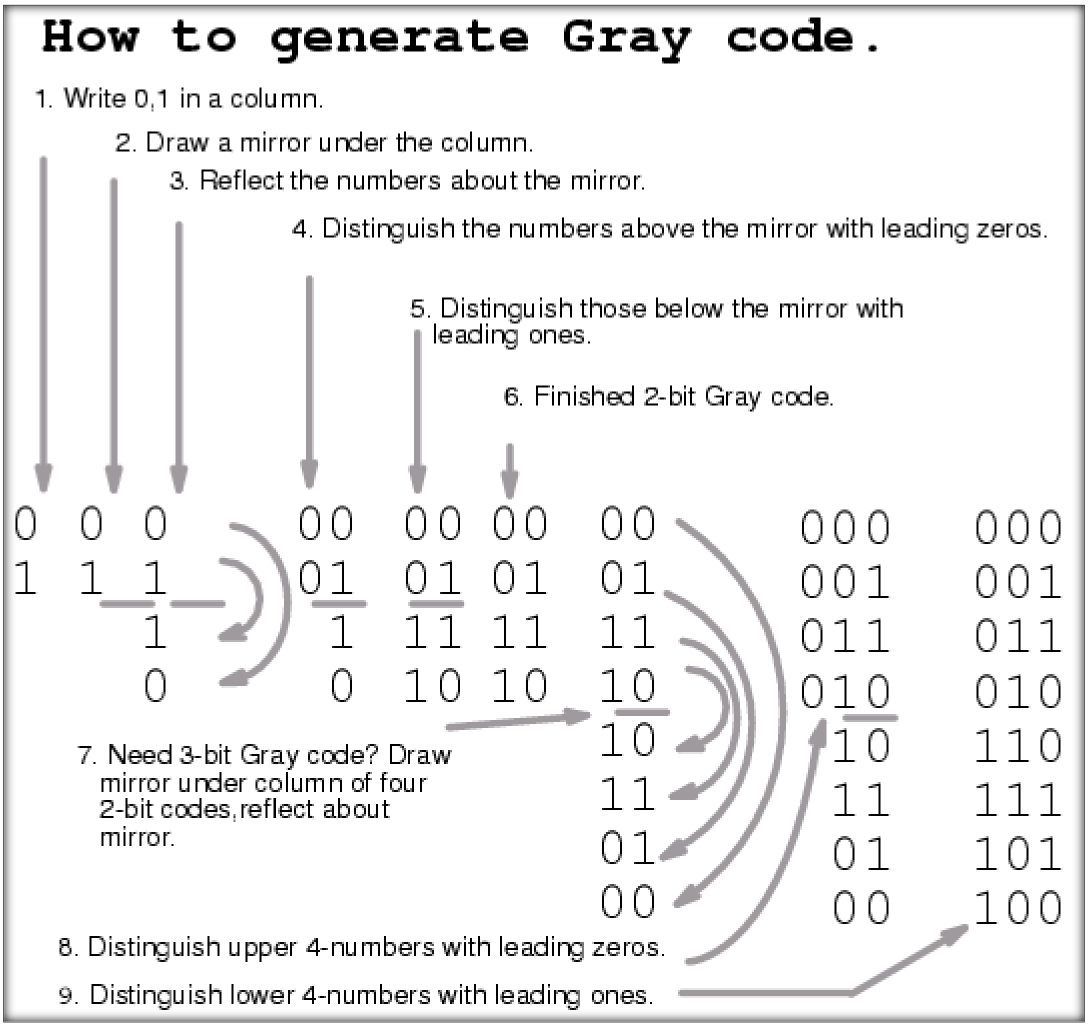
Theorem		Explanation
17.	$\overline{(A + B)} = \bar{A} \bullet \bar{B}$	For any two inputs A, B - the result of the inverse of A **or** B is the same as the inverse of A **and** the inverse of B.
18.	$\overline{(A \bullet B)} = \bar{A} + \bar{B}$	For any two inputs A, B - the result of the inverse of A **and** B is the same as the inverse of A **or** the inverse of B.

2.2. PENGENALAN PETA KARNAUGH

Peta karnaugh digunakan untuk mempermmudah penyederhanaan fungsi boolean baik untuk siku *minterm* atau *maxterm*. Gambar 1 merupakan contoh dari peta Karnaugh tiga variable. Yang perlu diperhatikan adalah deretan nomor pada kotak peta Karnaugh bukan diurutkan berdasarkan angka biner, melainkan berdasarkan deretan *GrayCode* seperti pada Gambar 2. Peta Karnaugh digambarkan dengan kotak bujur sangkar dengan tiap kotak merepresentasikan minterm (SoP). Jumlah kotak adalah dua pangkat jumlah variablenya. Pemahaman tentang peta Karnaugh akan dijelaskan lebih dalam pada modul 6.



Gambar 1. Contoh peta Karnaugh tiga variable (A, B, C)

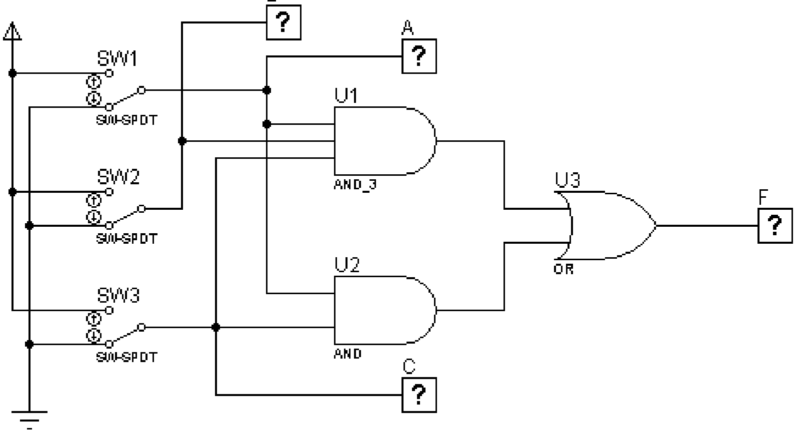


Gambar 2. Deretan *graycode*.

3. KEGIATAN PRAKTIKUM

Percobaan 1

1. Buat kombinasi gerbang logika sebagaimana pada gambar di bawah ini!



2. Fungsi boolean : **F = ABC + AC**

3. Isi titik-titik dalam tabel!

A	B	C	F
0	0	0	0
1	0	0	0
0	1	0	0
1	1	0	0
0	0	1	0
1	0	1	1
0	1	1	0
1	1	1	1

4. Isi titik-titik dalam karnaugh map

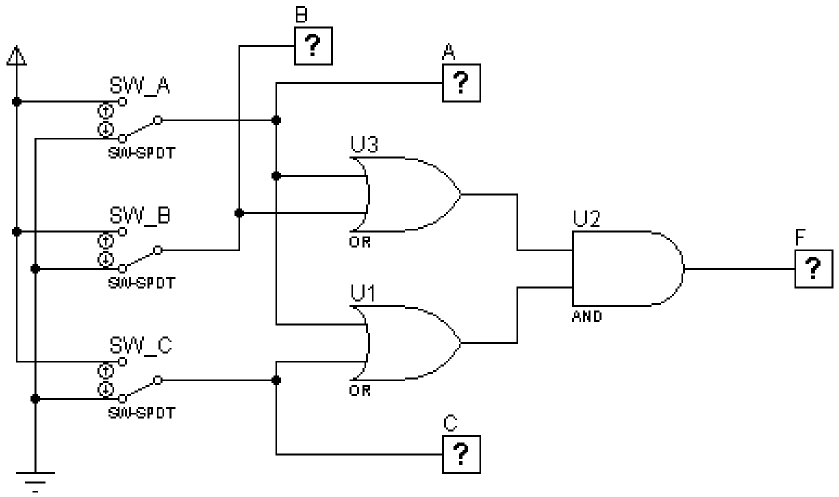
		AB			
		00	01	11	10
C	0	0	0	0	0
	1	0	0	1	1

5. Sederhanakan Fungsi boolean berdasarkan karnaugh map :
   F = AC.

6. Buat kombinasi gerbang logika berdasarkan fungsi boolean baru anda! gambar gerbang logika dalam kotak dibawah!

Percobaan 2

1. Buat kombinasi gerbang logika sebagaimana pada gambar!



2. Fungsi boolean : **F = (A + B). (A + C)**

3. Isi titik-titik dalam tabel kebenaran

A	B	C	F
0	0	0	0
1	0	0	1
0	1	0	0
1	1	0	1
0	0	1	0
1	0	1	1
0	1	1	1
1	1	1	1

4. Isi titik-titik dalam karnaugh map

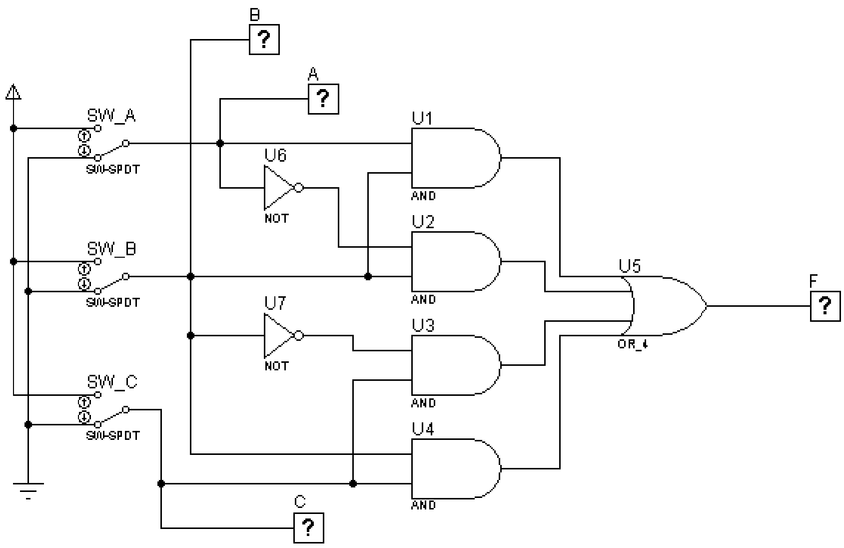
		AB			
		00	01	11	10
C	0	0	0	1	1
	1	0	1	1	1

5. Sederhanakan Fungsi boolean berdasarkan karnaugh map :
   F = ...

6. Buat kombinasi gerbang logika berdasarkan fungsi boolean baru anda! gambar gerbang logika dalam kotak dibawah!

Percobaan 3

1. Buat kombinasi gerbang logika sebagaimana pada gambar!



2. Fungsi boolean : **F = AB + A’B + B’C + BC**

3. Isi titik-titik dalam tabel kebenaran

A	B	C	F
0	0	0	0
1	0	0	0
0	1	0	1
1	1	0	1
0	0	1	1
1	0	1	1
0	1	1	1
1	1	1	1

4. Isi titik-titik dalam karnaugh map

		AB			
		00	01	11	10
C	0	0	1	1	0
	1	1	1	1	1

5. Sederhanakan Fungsi boolean berdasarkan karnaugh map :
F = ...

6. Buat kombinasi gerbang logika baru! Gambar dalam kotak dibawah ini!